Lecture 3

Myerson's Lemma

Last lecture advocated a two-step approach to the design of auctions that are DSIC, welfare maximizing, and computationally efficient (Section 2.6.4). The first step assumes truthful bids and identifies how to allocate items to bidders to maximize the social welfare. For instance, in sponsored search auctions, this step is implemented by assigning the ith highest bidder to the ith best slot. The second step derives the appropriate selling prices, to render truthful bidding a dominant strategy. This lecture states and proves *Myerson's lemma*, a powerful and general tool for implementing this second step. This lemma applies to sponsored search auctions as a special case, and Lectures 4 and 5 provide further applications.

Section 3.1 introduces single-parameter environments, a convenient generalization of the mechanism design problems introduced in Lecture 2. Section 3.2 rephrases the three steps of sealed-bid auctions (Section 2.2) in terms of allocation and payment rules. Section 3.3 defines two properties of allocation rules, implementability and monotonicity, and states and interprets Myerson's lemma. Section 3.4 gives a proof sketch of Myerson's lemma; it can be skipped on a first reading. Myerson's lemma includes a formula for payments in DSIC mechanisms, and Section 3.5 applies this formula to sponsored search auctions.

3.1 Single-Parameter Environments

A good level of abstraction at which to state Myerson's lemma is *single-parameter environments*. Such an environment has some number n of agents. Each agent i has a private nonnegative valuation v_i, her value "per unit of stuff" that she acquires. Finally, there is a *feasible set* X. Each element of X is a nonnegative n-vector $(x_1, x_2, \ldots, x_n)$, where x_i denotes the "amount of stuff" given

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

to agent i.

Example 3.1 (Single-Item Auction) In a single-item auction (Section 2.1), X is the set of 0-1 vectors that have at most one 1—that is, $\sum_{i=1}^{n} x_i \leq 1$.

Example 3.2 (k-Unit Auction) With k identical items and the constraint that each bidder gets at most one (Exercise 2.3), the feasible set is the set of 0-1 vectors that satisfy $\sum_{i=1}^{n} x_i \leq k$.

Example 3.3 (Sponsored Search Auction) In a sponsored search auction (Section 2.6), X is the set of n-vectors corresponding to assignments of bidders to slots, where each slot is assigned to at most one bidder and each bidder is assigned to at most one slot. If bidder i is assigned to slot j, then the component x_i equals the click-through rate α_j of her slot.

Example 3.4 (Public Project) Deciding whether or not to build a public project that can be used by all, such as a new bridge, can be modeled by the set $X = \{(0, 0, \ldots, 0), (1, 1, \ldots, 1)\}$.

Example 3.4 shows that single-parameter environments are general enough to capture applications different from auctions. At this level of generality, we refer to *agents* rather than bidders. We sometimes use the term *reports* instead of bids. A *mechanism* is a general procedure for making a decision when agents have private information (like valuations), whereas an *auction* is a mechanism specifically for the exchange of goods and money. See also Table 3.1.

auction	mechanism
bidder	agent
bid	report
valuation	valuation

Table 3.1: Correspondence of terms in auctions and mechanisms. An auction is the special case of a mechanism that is designed for the exchange of goods and money.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

3.2 Allocation and Payment Rules

Recall that a sealed-bid auction has to make two choices: who wins and who pays what. These two decisions are formalized via an *allocation rule* and a *payment rule*, respectively. Here are the three steps:

1. Collect bids $\mathbf{b} = (b_1, \ldots, b_n)$ from all agents. The vector $\mathbf{b}$ is called the *bid vector* or *bid profile*.
2. **[allocation rule]** Choose a feasible allocation $\mathbf{x}(\mathbf{b}) \in X \subseteq \mathbb{R}^n$ as a function of the bids.
3. **[payment rule]** Choose payments $\mathbf{p}(\mathbf{b}) \in \mathbb{R}^n$ as a function of the bids.

Procedures of this type are called *direct-revelation mechanisms*, because in the first step agents are asked to reveal directly their private valuations. An example of an indirect mechanism is an iterative ascending auction (cf., Exercise 2.7).

With our quasilinear utility model, in a mechanism with allocation and payment rules $\mathbf{x}$ and $\mathbf{p}$, respectively, agent i receives utility

$$u_i(\mathbf{b}) = v_i \cdot x_i(\mathbf{b}) - p_i(\mathbf{b})$$

when the bid profile is $\mathbf{b}$.

We focus on payment rules that satisfy

$$p_i(\mathbf{b}) \in [0, b_i \cdot x_i(\mathbf{b})] \tag{3.1}$$

for every agent i and bid profile $\mathbf{b}$. The constraint that $p_i(\mathbf{b}) \geq 0$ is equivalent to prohibiting the seller from paying the agents. The constraint that $p_i(\mathbf{b}) \leq b_i \cdot x_i(\mathbf{b})$ ensures that a truthful agent receives nonnegative utility (do you see why?).[1]

3.3 Statement of Myerson's Lemma

Next are two important definitions. Both articulate a property of allocation rules.

[1] There are applications where it makes sense to relax one or both of these restrictions on payments, including those discussed in Exercise 2.5 and Problem 7.1.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

Definition 3.5 (Implementable Allocation Rule) An allocation rule $\mathbf{x}$ for a single-parameter environment is *implementable* if there is a payment rule $\mathbf{p}$ such that the direct-revelation mechanism $(\mathbf{x}, \mathbf{p})$ is DSIC.

That is, the implementable allocation rules are those that extend to DSIC mechanisms. Equivalently, the projection of DSIC mechanisms onto their allocation rules is the set of implementable rules. If our aim is to design a DSIC mechanism, we must confine ourselves to implementable allocation rules—they form our "design space." In this terminology, we can rephrase the cliffhanger from the end of Lecture 2 as: is the welfare-maximizing allocation rule for sponsored search auctions, which assigns the ith highest bidder to the ith best slot, implementable?

For instance, consider a single-item auction (Example 3.1). Is the allocation rule that awards the item to the highest bidder implementable? Sure—we've already constructed a payment rule, the second-price rule, that renders it DSIC. What about the allocation rule that awards the item to the *second-highest* bidder? Here, the answer is not clear: we haven't seen a payment rule that extends it to a DSIC mechanism, but it also seems tricky to argue that no payment rule could conceivably work.

Definition 3.6 (Monotone Allocation Rule) An allocation rule $\mathbf{x}$ for a single-parameter environment is *monotone* if for every agent i and bids $\mathbf{b}_{-i}$ by the other agents, the allocation $x_i(z, \mathbf{b}_{-i})$ to i is nondecreasing in her bid z.

That is, in a monotone allocation rule, bidding higher can only get you more stuff.

For example, the single-item auction allocation rule that awards the item to the highest bidder is monotone: if you're the winner and you raise your bid (keeping other bids fixed), you continue to win. By contrast, awarding the item to the second-highest bidder is a non-monotone allocation rule: if you're the winner and you raise your bid high enough, you lose.

The welfare-maximizing allocation rule for sponsored search auctions (Example 3.3), with the ith highest bidder awarded the ith best slot, is monotone. When a bidder raises her bid, her position in the

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

sorted order of bids can only increase, and this can only increase the click-through rate of her assigned slot.

We state Myerson's lemma in three parts; each is conceptually interesting and useful in later applications.

Theorem 3.7 (Myerson's Lemma) *Fix a single-parameter environment.*

(a) An allocation rule $\mathbf{x}$ *is implementable if and only if it is monotone.*

(b) If $\mathbf{x}$ *is monotone, then there is a unique payment rule for which the direct-revelation mechanism* $(\mathbf{x}, \mathbf{p})$ *is DSIC and* $p_i(\mathbf{b}) = 0$ *whenever* $b_i = 0$.

(c) The payment rule in (b) is given by an explicit formula.[2]

Myerson's lemma is the foundation on which we'll build most of our mechanism design theory. Part (a) states that Definitions 3.5 and 3.6 define exactly the same class of allocation rules. This equivalence is incredibly powerful: Definition 3.5 describes our design goal but is unwieldy to work with and verify, while Definition 3.6 is far more "operational." Usually, it's not difficult to check whether or not an allocation rule is monotone. Part (b) states that when an allocation rule is implementable, there is no ambiguity in how to assign payments to achieve the DSIC property—there is only one way to do it. Moreover, there is a relatively simple and explicit formula for this payment rule (part (c)), a property we apply to sponsored search auctions in Section 3.5 and to revenue-maximizing auction design in Lectures 5–6.

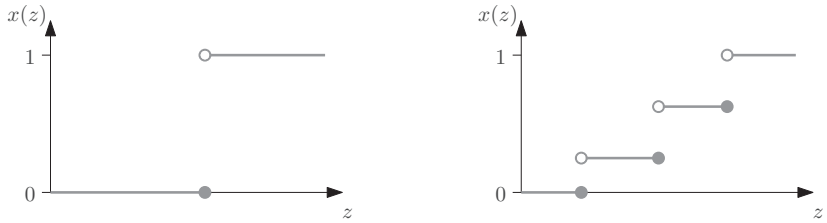
*3.4 Proof of Myerson's Lemma (Theorem 3.7)

Fix a single-parameter environment and consider an allocation rule $\mathbf{x}$, which may or may not be monotone. Suppose there is a payment rule $\mathbf{p}$ such that $(\mathbf{x}, \mathbf{p})$ is a DSIC mechanism—what could $\mathbf{p}$ look like? The plan of the proof is to use the stringent DSIC constraint to whittle the possibilities for $\mathbf{p}$ down to a single candidate. We establish all three parts of the theorem in one fell swoop.

[2] See formulas (3.5) and (3.6) for details and Section 3.5 for concrete examples.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

Recall the DSIC condition: for every agent i, every possible private valuation v_i, every set of bids $\mathbf{b}_{-i}$ by the other agents, it must be that i's utility is maximized by bidding truthfully. For now, fix i and $\mathbf{b}_{-i}$ arbitrarily. As shorthand, write $x(z)$ and $p(z)$ for the allocation $x_i(z, \mathbf{b}_{-i})$ and payment $p_i(z, \mathbf{b}_{-i})$ of i when she bids z, respectively. Figure 3.1 gives two examples of a possible *allocation curve*, meaning the graph of such an x as a function of z.



(a) 0-1 monotone curve

(b) Piecewise constant monotone curve

Figure 3.1: Examples of allocation curves $x(\cdot)$.

We invoke the DSIC constraint via a simple but clever swapping trick. Suppose $(\mathbf{x}, \mathbf{p})$ is DSIC, and consider any $0 \leq y < z$. Because agent i might well have private valuation z and is free to submit the false bid y, DSIC demands that

$$\underbrace{z \cdot x(z) - p(z)}_{\text{utility of bidding } z} \geq \underbrace{z \cdot x(y) - p(y).}_{\text{utility of bidding } y} \tag{3.2}$$

Similarly, since agent i might well have the private valuation y and could submit the false bid z, $(\mathbf{x}, \mathbf{p})$ must satisfy

$$\underbrace{y \cdot x(y) - p(y)}_{\text{utility of bidding } y} \geq \underbrace{y \cdot x(z) - p(z).}_{\text{utility of bidding } z} \tag{3.3}$$

Myerson's lemma is, in effect, trying to solve for the payment rule $\mathbf{p}$ given the allocation rule $\mathbf{x}$. Rearranging inequalities (3.2) and (3.3) yields the following "payment difference sandwich," bounding $p(y) - p(z)$ from below and above:

$$z \cdot [x(y) - x(z)] \leq p(y) - p(z) \leq y \cdot [x(y) - x(z)]. \tag{3.4}$$

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

The payment difference sandwich already implies that every implementable allocation rule is monotone (Exercise 3.1). Thus, we can assume for the rest of the proof that $\mathbf{x}$ is monotone.

Next, consider the case where x is a piecewise constant function, as in Figure 3.1. The graph of x is flat except for a finite number of "jumps." In (3.4), fix z and let y tend to z from above. Taking the limit $y \downarrow z$ from above in (3.4), the left- and right-hand sides become 0 if there is no jump in x at z. If there is a jump of magnitude h at z, then the left- and right-hand sides both tend to $z \cdot h$. This implies the following constraint on p, for every z:

$$\text{jump in } p \text{ at } z = z \cdot [\text{jump in } x \text{ at } z].$$

Combining this with the initial condition $p(0) = 0$, we've derived the following *payment formula*, for every agent i, bids $\mathbf{b}_{-i}$ by other agents, and bid b_i by i:

$$p_i(b_i, \mathbf{b}_{-i}) = \sum_{j=1}^{\ell} z_j \cdot [\text{jump in } x_i(\cdot, \mathbf{b}_{-i}) \text{ at } z_j], \tag{3.5}$$

where $z_1, \ldots, z_\ell$ are the breakpoints of the allocation function $x_i(\cdot, \mathbf{b}_{-i})$ in the range $[0, b_i]$.

A similar argument applies when x is a monotone function that is not piecewise constant. For instance, suppose that x is differentiable.[3] Dividing the payment difference sandwich (3.4) by $y - z$ and taking the limit as $y \downarrow z$ yields the constraint

$$p'(z) = z \cdot x'(z).$$

Combining this with the condition $p(0) = 0$ yields the payment formula

$$p_i(b_i, \mathbf{b}_{-i}) = \int_0^{b_i} z \cdot \tfrac{d}{dz} x_i(z, \mathbf{b}_{-i}) dz \tag{3.6}$$

for every agent i, bid b_i, and bids $\mathbf{b}_{-i}$ by the other agents.

We reiterate that these payment formulas give the *only possible* payment rule that has a chance of extending the given allocation rule $\mathbf{x}$ into a DSIC mechanism. Thus, for every allocation rule $\mathbf{x}$,

[3] With some additional facts from calculus, the proof extends to general monotone functions. We omit the details.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

there is at most one payment rule $\mathbf{p}$ such that $(\mathbf{x}, \mathbf{p})$ is DSIC (cf., part (b) of Theorem 3.7). But the proof is not complete: we still have to check that this payment rule works provided $\mathbf{x}$ is monotone! Indeed, we already know that even this payment rule cannot extend a non-monotone allocation rule to a DSIC mechanism.

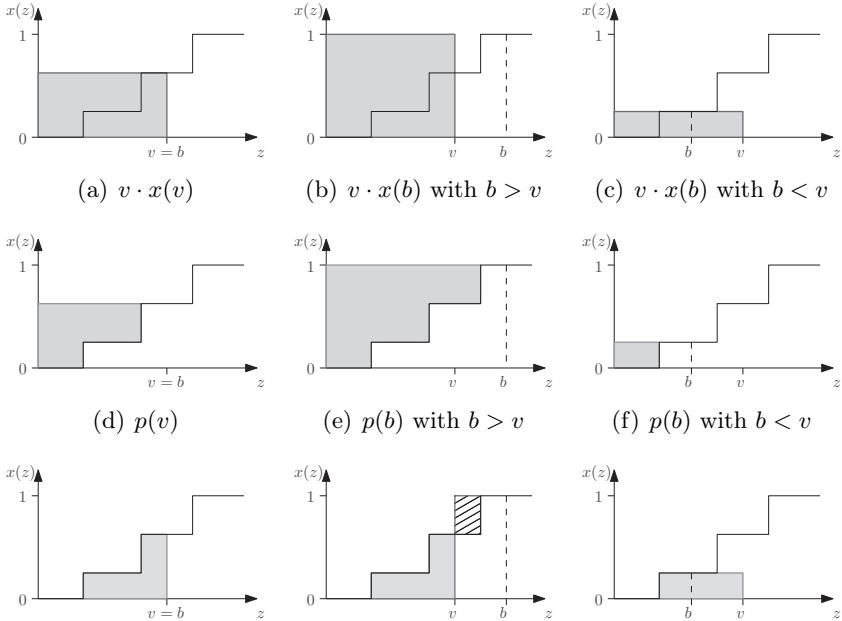
We give a proof by picture that, if $\mathbf{x}$ is a monotone and piecewise constant allocation rule and $\mathbf{p}$ is defined by (3.5), then $(\mathbf{x}, \mathbf{p})$ is a DSIC mechanism. The same argument works more generally for monotone allocation rules that are not piecewise constant, with payments defined as in (3.6). This will complete the proof of all three parts of Myerson's lemma.

Figures 3.2(a)–(i) depict the utility of an agent when she bids truthfully, overbids, and underbids, respectively. The allocation curve $x(z)$ and the private valuation v of the agent is the same in all three cases. Recall that the agent's utility when she bids b is $v \cdot x(b) - p(b)$. We depict the first term $v \cdot x(b)$ as a shaded rectangle of width v and height $x(b)$ (Figures 3.2(a)–(c)). Using the formula (3.5), we see that the payment $p(b)$ can be represented as the shaded area to the left of the allocation curve in the range $[0, b]$ (Figures 3.2(d)–(f)). The agent's utility is the difference between these two terms (Figures 3.2(g)–(i)). When the agent bids truthfully, her utility is precisely the area under the allocation curve in the range $[0, v]$ (Figure 3.2(g)). When the agent overbids, her utility is this same area, minus the area above the allocation curve in the range $[v, b]$ (Figure 3.2(h)). When the agent underbids, her utility is a subset of the area under the allocation curve in the range $[0, v]$ (Figure 3.2(i)). Since the agent's utility is the largest in the first case, the proof is complete.

3.5 Applying the Payment Formula

The explicit payment rule given by Myerson's lemma (Theorem 3.7(c)) is easy to understand and apply in many applications. For starters, consider a single-item auction (Example 3.1) with the allocation rule that allocates the item to the highest bidder. Fixing a bidder i and bids $\mathbf{b}_{-i}$ by the others, the function $x_i(z, \mathbf{b}_{-i})$ is 0 up to $B = \max_{j \neq i} b_j$ and 1 thereafter. Myerson's payment formula for such piecewise constant functions, derived in Section 3.4 as equation (3.5),

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004



(a) $v \cdot x(v)$ (b) $v \cdot x(b)$ with $b > v$ (c) $v \cdot x(b)$ with $b < v$

(d) $p(v)$ (e) $p(b)$ with $b > v$ (f) $p(b)$ with $b < v$

(g) utility with $b = v$ (h) utility with $b > v$ (i) utility with $b < v$

Figure 3.2: Proof by picture that the payment rule in (3.5), coupled with the given monotone and piecewise constant allocation rule, yields a DSIC mechanism. The three columns consider the cases of truthful bidding, overbidding, and underbidding, respectively. The three rows show the welfare $v \cdot x(b)$, the payment $p(b)$, and the utility $v \cdot x(b) - p(b)$, respectively. In (h), the solid region represents positive utility and the lined region represents negative utility.

is

$$p_i(b_i, \mathbf{b}_{-i}) = \sum_{j=1}^{\ell} z_j \cdot [\text{jump in } x_i(\cdot, \mathbf{b}_{-i}) \text{ at } z_j],$$

where $z_1, \ldots, z_\ell$ are the breakpoints of the allocation function $x_i(\cdot, \mathbf{b}_{-i})$ in the range $[0, b_i]$. For the highest-bidder single-item allocation rule, this is either 0 (if $b_i < B$) or, if $b_i > B$, there is a single breakpoint (a jump of 1 at B) and the payment is $p_i(b_i, \mathbf{b}_{-i}) = B$. Thus, Myerson's lemma regenerates the second-price payment rule as a special case.

Next, consider a sponsored search auction (Example 3.3), with

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

k slots with click-through rates $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_k$. Let $\mathbf{x}(\mathbf{b})$ be the allocation rule that assigns the ith highest bidder to the ith best slot, for $i = 1, 2, \ldots, k$. This rule is monotone and, assuming truthful bids, welfare maximizing (Exercise 2.8). Myerson's payment formula then gives the unique payment rule $\mathbf{p}$ such that $(\mathbf{x}, \mathbf{p})$ is DSIC. To describe it, consider a bid profile $\mathbf{b}$, and re-index the bidders so that $b_1 \geq b_2 \geq \cdots \geq b_n$. First focus on the highest bidder and imagine increasing her bid from 0 to b_1, holding the other bids fixed. The allocation $x_1(z, \mathbf{b}_{-1})$ increases from 0 to α_1 as z increases from 0 to b_1, with a jump of $\alpha_j - \alpha_{j+1}$ at the point where z becomes the jth highest bid in the profile $(z, \mathbf{b}_{-1})$, namely b_{j+1}. In general, Myerson's payment formula gives the payment

$$p_i(\mathbf{b}) = \sum_{j=i}^{k} b_{j+1}(\alpha_j - \alpha_{j+1}) \tag{3.7}$$

for the ith highest bidder (where $\alpha_{k+1} = 0$).

Our sponsored search model assumes that bidders don't care about impressions (i.e., having their link shown), except inasmuch as it leads to a click. This motivates charging bidders per click rather than per impression. The per-click payment for the bidder i in slot i is simply that in (3.7), scaled by $\frac{1}{\alpha_i}$:

$$p_i(\mathbf{b}) = \sum_{j=i}^{k} b_{j+1}\frac{\alpha_j - \alpha_{j+1}}{\alpha_i}. \tag{3.8}$$

Thus, when a bidder's link is clicked, she pays a suitable convex combination of the lower bids.

By historical accident, the sponsored search auctions used by search engines are based on the "generalized second-price (GSP)" auction, which is a simpler version of the DSIC auction above. The GSP allocation rule also assigns the ith highest bidder to the ith best slot, but its payment rule charges this bidder a per-click payment equal to the $(i+1)$th highest bid. For a fixed set of bids, these per-click payments are generally higher than those in (3.8). Myerson's lemma asserts that the payment rule in (3.8) is the unique one that, when coupled with the GSP allocation rule, yields a DSIC mechanism. We can immediately conclude that the GSP auction is *not* DSIC. It still has a number of nice properties, however, and is equivalent to the

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

DSIC auction in certain senses. Problem 3.1 explores this point in detail.

The Upshot

- ☆ In a single-parameter environment, every agent has a private valuation per "unit of stuff," and a feasible set defines how much "stuff" can be jointly allocated to the agents. Examples include single-item auctions, sponsored search auctions, and public projects.
- ☆ The allocation rule of a direct-revelation mechanism specifies, as a function of agents' bids, who gets what. The payment rule of such a mechanism specifies, as a function of agents' bids, who pays what.
- ☆ An allocation rule is implementable if there exists a payment rule that extends it to a DSIC mechanism.
- ☆ An allocation rule is monotone if bidding higher can only increase the amount of stuff allocated to an agent, holding other agents' bids fixed.
- ☆ Myerson's lemma states that an allocation rule is implementable if and only if it is monotone. In this case, the corresponding payment rule is unique (assuming that bidding 0 results in paying 0).
- ☆ There is an explicit formula, given in (3.5) and (3.6), for the payment rule that extends a monotone allocation rule to a DSIC mechanism.
- ☆ Myerson's payment formula yields an elegant payment rule (3.8) for the payments-per-click in an ideal sponsored search auction.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

Notes

Myerson's lemma is from Myerson (1981). The sponsored search payment formula (3.8) is noted in Aggarwal et al. (2006). Problem 3.1 is due independently to Edelman et al. (2007) and Varian (2007). Problem 3.2 is related to the "profit extractors" introduced by Goldberg et al. (2006) and the cost-sharing mechanisms studied by Moulin and Shenker (2001). Problem 3.3 is a special case of the theory developed by Moulin and Shenker (2001).

Exercises

Exercise 3.1 Use the "payment difference sandwich" in (3.4) to prove that if an allocation rule is not monotone, then it is not implementable.

Exercise 3.2 The proof of Myerson's lemma (Section 3.4) concludes with a "proof by picture" that coupling a monotone and piecewise constant allocation rule $\mathbf{x}$ with the payment rule defined by (3.5) yields a DSIC mechanism. Where does the proof-by-picture break down if the piecewise constant allocation rule $\mathbf{x}$ is not monotone?

Exercise 3.3 Give an algebraic proof that coupling a monotone and piecewise constant allocation rule $\mathbf{x}$ with the payment rule defined by (3.5) yields a DSIC mechanism.

Exercise 3.4 Consider the following extension of the sponsored search setting described in Section 2.6. Each bidder i now has a publicly known *quality* β_i, in addition to a private valuation v_i per click. As usual, each slot j has a CTR α_j, and $\alpha_1 \geq \alpha_2 \cdots \geq \alpha_k$. We assume that if bidder i is placed in slot j, then the probability of a click is $\beta_i \alpha_j$. Thus bidder i derives value $v_i \beta_i \alpha_j$ from the jth slot.

Describe the welfare-maximizing allocation rule in this generalized sponsored search setting. Prove that this rule is monotone. Give an explicit formula for the per-click payment of each bidder that extends this allocation rule to a DSIC mechanism.

Problems

Problem 3.1 Recall our model of sponsored search auctions (Section 2.6): there are k slots, the jth slot has a click-through rate

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

(CTR) of α_j (nonincreasing in j), and the utility of bidder i in slot j is $\alpha_j(v_i - p_j)$, where v_i is the (private) value-per-click of the bidder and p_j is the price charged per-click in slot j. The *Generalized Second-Price (GSP)* auction is defined as follows:

The Generalized Second Price (GSP) Auction

1. Rank advertisers from highest to lowest bid; assume without loss of generality that $b_1 \geq b_2 \geq \cdots \geq b_n$.
2. For $i = 1, 2, \ldots, k$, assign the ith bidder to the i slot.
3. For $i = 1, 2, \ldots, k$, charge the ith bidder a price of b_{i+1} per click.

The following subproblems show that the GSP auction always has a canonical equilibrium that is equivalent to the truthful outcome in the corresponding DSIC sponsored search auction.

(a) Prove that for every $k \geq 2$ and sequence $\alpha_1 \geq \cdots \geq \alpha_k > 0$ of CTRs, the GSP auction is not DSIC.

(b) *(H)* Fix CTRs for slots and values-per-click for bidders. We can assume that $k = n$ by adding dummy slots with zero CTR (if $k < n$) or dummy bidders with zero value-per-click (if $k > n$). A bid profile $\mathbf{b}$ is an *equilibrium* of GSP if no bidder can increase her utility by unilaterally changing her bid. Verify that this condition translates to the following inequalities, under our standing assumption that $b_1 \geq b_2 \geq \cdots \geq b_n$: for every i and higher slot $j < i$,

$$\alpha_i(v_i - b_{i+1}) \geq \alpha_j(v_i - b_j);$$

and for every lower slot $j > i$,

$$\alpha_i(v_i - b_{i+1}) \geq \alpha_j(v_i - b_{j+1}).$$

(c) A bid profile $\mathbf{b}$ with $b_1 \geq \cdots \geq b_n$ is *envy-free* if for every bidder i and slot $j \neq i$,

$$\alpha_i(v_i - b_{i+1}) \geq \alpha_j(v_i - b_{j+1}). \tag{3.9}$$

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

Verify that every envy-free bid profile is an equilibrium.[4]

(d) *(H)* A bid profile is *locally envy-free* if the inequality (3.9) holds for every pair of adjacent slots—for every i and $j \in \{i-1, i+1\}$. By definition, an envy-free bid profile is also locally envy-free. Prove that, for every set of strictly decreasing CTRs, every locally envy-free bid profile is also envy-free.

(e) *(H)* Prove that, for every set of values-per-click and strictly decreasing CTRs, there is an equilibrium of the GSP auction in which the assignments of bidders to slots and all payments-per-click equal those in the truthful outcome of the corresponding DSIC sponsored search auction.

(f) Prove that the equilibrium in (e) is the lowest-revenue envy-free bid profile.

Problem 3.2 This problem considers a k-unit auction (Example 3.2) in which the seller has a specific revenue target R. Consider the following algorithm that, given bids $\mathbf{b}$ as input, determines the winning bidders and their payments.

Revenue Target Auction

initialize a set S to the top k bidders
while there is a bidder $i \in S$ with $b_i < R/|S|$ **do**
    remove an arbitrary such bidder from S
allocate an item to each bidder of S (if any) at a price of $R/|S|$

(a) Give an explicit description of the allocation rule of the Revenue Target Auction, and prove that it is monotone.

(b) *(H)* Conclude from Myerson's lemma that the Revenue Target Auction is a DSIC mechanism.

[4] Why "envy-free?" Because if we write $p_j = b_{j+1}$ for the current price-per-click of slot j, then these inequalities translate to: "every bidder i is as happy getting her current slot at her current price as she would be getting any other slot and that slot's current price."

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

(c) Prove that whenever the DSIC and welfare-maximizing k-unit auction (Exercise 2.3) obtains revenue at least R, the Revenue Target Auction obtains revenue R.

(d) Prove that there exists a valuation profile for which the Revenue Target Auction obtains revenue R but the DSIC and welfare-maximizing auction earns revenue less than R.

Problem 3.3 This problem revisits the issue of collusion in auctions; see also Problem 2.2.

(a) Prove that the Revenue Target Auction in Problem 3.2 is *group-strategyproof*, meaning that no coordinated false bids by a subset of bidders can ever strictly increase the utility of one of its members without strictly decreasing the utility of some other member.

(b) Is the DSIC and welfare-maximizing k-unit auction group-strategyproof?

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:21, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.004

Lecture 4

Algorithmic Mechanism Design

This lecture pursues mechanisms that are DSIC, welfare maximizing, and computationally efficient for single-parameter environments that are more complex than those in Lectures 2 and 3. These environments are general enough that the welfare maximization problem is $\mathcal{NP}$-hard, so we consider allocation rules that only maximize the social welfare approximately. There are many techniques for designing such rules, but not all of them yield rules that are monotone in the sense required by Myerson's lemma. This lecture also discusses the revelation principle, the formal justification for our restriction to direct-revelation mechanisms.

Section 4.1 introduces knapsack auctions, which are conceptually simple single-parameter environments in which welfare maximization is a computationally intractable (i.e., $\mathcal{NP}$-hard) problem. Section 4.2 uses knapsack auctions to illustrate some representative results in algorithmic mechanism design, where the goal to design DSIC and polynomial-time mechanisms that guarantee near-optimal welfare. Section 4.3 presents the revelation principle.

4.1 Knapsack Auctions

4.1.1 Problem Definition

Knapsack auctions are another example of single-parameter environments (Section 3.1).

Example 4.1 (Knapsack Auction) In a knapsack auction, each bidder i has a publicly known *size* w_i and a private valuation. The seller has a capacity W. The feasible set X is defined as the 0-1 vectors $(x_1, \ldots, x_n)$ such that $\sum_{i=1}^{n} w_i x_i \leq W$. (As usual, $x_i = 1$ indicates that i is a winning bidder.)

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

Whenever there is a shared resource with limited capacity, you have a knapsack problem. For instance, each bidder's size could represent the duration of a company's television ad, the valuation its willingness-to-pay for its ad being shown during the Super Bowl, and the seller capacity the length of a commercial break. Other situations that can be modeled with knapsack auctions include bidders who want files stored on a shared server, data streams sent through a shared communication channel, or processes to be executed on a shared supercomputer. A k-unit auction (Example 3.2) corresponds to a knapsack auction with $w_i = 1$ for all i and $W = k$.

Let's try to design an ideal auction using our two-step design paradigm (Section 2.6.4). First, we assume without justification that we receive truthful bids and decide on our allocation rule. Then we pay the piper and devise a payment rule that extends the allocation rule to a DSIC mechanism.

4.1.2 Welfare-Maximizing DSIC Knapsack Auctions

Since ideal auctions are supposed to maximize welfare, the answer to the first step is clear: define the allocation rule by

$$\mathbf{x}(\mathbf{b}) = \text{argmax}_X \sum_{i=1}^{n} b_i x_i. \tag{4.1}$$

That is, the allocation rule solves an instance of the knapsack problem[1] in which the item (i.e., bidder) values are the reported bids $b_1, \ldots, b_n$, and the item sizes are the a priori known sizes $w_1, \ldots, w_n$. By definition, when bidders bid truthfully, this allocation rule maximizes the social welfare. This allocation rule is also monotone in the sense of Definition 3.6; see Exercise 4.1.

4.1.3 Critical Bids

Myerson's lemma (Theorem 3.7, parts (a) and (b)) guarantees the existence of a payment rule $\mathbf{p}$ such that the mechanism $(\mathbf{x}, \mathbf{p})$ is DSIC. This payment rule is easy to understand. Fix a bidder i and

[1] An instance of the knapsack problem is defined by $2n + 1$ positive numbers: item values $v_1, \ldots, v_n$, item sizes $w_1, \ldots, w_n$, and a knapsack capacity W. The goal is to compute the subset of items of maximum total value that has total size at most W.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

bids $\mathbf{b}_{-i}$ by the other bidders. Since the allocation rule is monotone and assigns 0 or 1 to every bidder, the allocation curve $x_i(\cdot, \mathbf{b}_{-i})$ is 0 until some breakpoint z, at which point it jumps to 1 (Figure 4.1). Recall the payment formula in (3.5):

$$p_i(b_i, \mathbf{b}_{-i}) = \sum_{j=1}^{\ell} z_j \cdot [\text{jump in } x_i(\cdot, \mathbf{b}_{-i}) \text{ at } z_j],$$

where $z_1, \ldots, z_\ell$ are the breakpoints of the allocation function $x_i(\cdot, \mathbf{b}_{-i})$ in the range $[0, b_i]$. Thus, if i bids less than z, she loses and pays 0. If i bids more than z, she pays $z \cdot (1 - 0) = z$. That is, when i wins, she pays her *critical bid*—the infimum of the bids she could make and continue to win (holding the other bids $\mathbf{b}_{-i}$ fixed). This is analogous to the familiar payment rule of a single-item second-price auction.

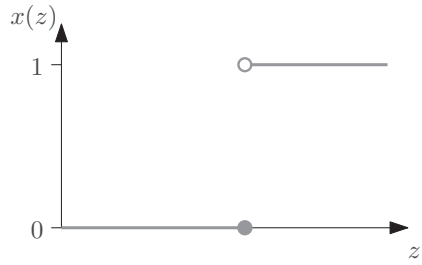


Figure 4.1: A monotone 0-1 allocation rule.

4.1.4 Intractability of Welfare Maximization

Is the mechanism proposed in Section 4.1.2 ideal in the sense of the second-price auction (Theorem 2.4)? Recall this means that the mechanism:

(1) is DSIC;

(2) is welfare maximizing, assuming truthful bids; and

(3) runs in time polynomial in the input size, which is the number of bits needed to represent all of the relevant numbers (bids, sizes, and the capacity).[2]

[2] More precisely, the running time of the mechanism on inputs of size s should be at most cs^d, where c and d are constants independent of s.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

The answer is *no*. The reason is that the knapsack problem is $\mathcal{NP}$-hard. This means that there is no polynomial-time implementation of the allocation rule in (4.1), unless $\mathcal{P} = \mathcal{NP}$.[3] Thus, properties (2) and (3) are incompatible.

The fact that there is no ideal knapsack auction (assuming $\mathcal{P} \neq \mathcal{NP}$) motivates relaxing at least one of our three goals. But which one? First, note that relaxing the DSIC condition will not help at all, since it is the second and third properties that conflict.

One sensible approach, which won't get much airtime in this course, is to relax the third constraint. This is particularly attractive for knapsack auctions, since the allocation rule (4.1) can be implemented in pseudopolynomial time using dynamic programming.[4] More generally in mechanism design, if your instances are small or structured enough and you have enough time and computing power to implement optimal welfare maximization, by all means do it. The resulting allocation rule is monotone and can be extended to a DSIC mechanism (Exercise 4.1).[5]

For the rest of this lecture, we'll compromise on the second goal—begrudgingly accepting near-optimal welfare maximization in exchange for computational efficiency and without losing DSIC.

4.2 Algorithmic Mechanism Design

Algorithmic mechanism design is one of the first and most well-studied branches of algorithmic game theory, and this section presents a representative result from the field.

The dominant paradigm in algorithmic mechanism design is to relax the second requirement of ideal auctions (welfare maximization) as little as possible, subject to the first (DSIC) and third (polynomial-time) requirements. For single-parameter environments, Myerson's

[3] $\mathcal{P}$ and $\mathcal{NP}$ denote the sets of problems that can be solved in polynomial time and for which a correct solution can be verified in polynomial time, respectively. $\mathcal{NP}$ can only be larger than $\mathcal{P}$, and $\mathcal{P} \neq \mathcal{NP}$ is an unproven but widely-accepted hypothesis about computation.

[4] That is, if either the bids or the sizes require only a small number of bits to describe, then the problem can be solved efficiently. See any undergraduate algorithms textbook for details.

[5] Don't forget that the payments also need to be computed, and this generally requires solving n more welfare maximization problems (one per agent). See also Exercise 4.3.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

lemma (Theorem 3.7) reduces this task to the design of a polynomial-time and monotone allocation rule that comes as close as possible to maximizing the social welfare.

4.2.1 The Best-Case Scenario: DSIC for Free

One reason there has been so much progress in algorithmic mechanism design over the past 15 years is its strong resemblance to the mature field of *approximation algorithms*. The primary goal in approximation algorithms is to design polynomial-time algorithms for $\mathcal{NP}$-hard optimization problems that are as close to optimal as possible. Algorithmic mechanism design has exactly the same goal, except that the algorithms must additionally obey a monotonicity constraint. The incentive constraints of the mechanism design goal are neatly compiled into a relatively intuitive extra constraint on the allocation rule, and so algorithmic mechanism design reduces to algorithm design in an oddly restricted "computational model."

The design space of polynomial-time DSIC mechanisms is only smaller than that of polynomial-time approximation algorithms. The best-case scenario is that the extra DSIC (equivalently, monotonicity) constraint causes no additional welfare loss, beyond the loss we already have to suffer from the polynomial-time requirement. We've been spoiled so far, since exact welfare maximization automatically yields a monotone allocation rule (see Exercise 4.1). Does an analogous fact hold for *approximate* welfare maximization?

4.2.2 Knapsack Auctions Revisited

To explore the preceding question in a concrete setting, let's return to knapsack auctions. There are several heuristics for the knapsack problem that have good worst-case performance guarantees. For example, consider the following allocation rule, which given bids $\mathbf{b}$ chooses a feasible set—a set S of winners with total size $\sum_{i \in S} w_i$ at most the capacity W—via a simple greedy algorithm. Since it's harmless to remove bidders i with $w_i > W$, we can assume that $w_i \leq W$ for every i.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

A Greedy Knapsack Heuristic

1. Sort and re-index the bidders so that
$$\frac{b_1}{w_1} \geq \frac{b_2}{w_2} \geq \cdots \geq \frac{b_n}{w_n}.$$[6]
2. Pick winners in this order until one doesn't fit, and then halt.[7]
3. Return either the solution from the previous step or the highest bidder, whichever has larger social welfare.[8]

This greedy algorithm is a $\frac{1}{2}$-approximation algorithm for the knapsack problem, meaning that for every instance of the knapsack problem, the algorithm returns a feasible solution with total value at least $\frac{1}{2}$ times the maximum possible. This fact implies the following guarantee.

Theorem 4.2 (Knapsack Approximation Guarantee) *Assuming truthful bids, the social welfare achieved by the greedy allocation rule is at least 50% of the maximum social welfare.*

Proof (sketch): Consider truthful bids $v_1, \ldots, v_n$, known sizes w_1, $\ldots$, w_n, and a capacity W. Suppose, as a thought experiment, we make the problem easier by allowing bidders to be chosen *fractionally*, with the value prorated accordingly. For example, if 70% of a bidder with value 10 is chosen, then it contributes 7 to the welfare. Here is a greedy algorithm for this "fractional knapsack problem:" sort the bidders as in step (1) above, and pick winners in this order until the entire capacity is fully used (picking the final winner fractionally,

[6] Intuitively, what makes a bidder attractive is a high bid and a small size. This heuristic trades these two properties off by ordering bidders by "bang-per-buck"—the value contributed per unit of capacity used.

[7] Alternatively, continue to follow the sorted order, packing any further bidders that fit. This modified heuristic is only better than the original.

[8] The motivation for this step is that the solution produced by the second step can be highly suboptimal if there is a very valuable and very large bidder. In lieu of considering only the highest bidder, this step can also sort the bidders in nondecreasing bid order and pack them greedily. This modification can only improve the heuristic.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

as needed). A straightforward exchange argument proves that this algorithm maximizes the welfare over all feasible solutions to the fractional knapsack problem (Exercise 4.4).

In the optimal fractional solution, suppose that the first k bidders in the sorted order win and that the $(k+1)$th bidder fractionally wins. The welfare achieved by steps (1) and (2) in the greedy allocation rule is exactly the total value of the first k bidders. The welfare of the solution consisting only the highest bidder is at least the fractional value of the $(k+1)$th bidder. The better of these two solutions is at least half of the welfare of the optimal fractional solution, and thus at least half the welfare of an optimal (non-fractional) solution to the original problem. ■

The greedy allocation rule above is also monotone (Exercise 4.5). Using Myerson's lemma (Theorem 3.7), we can extend it to a DSIC mechanism that runs in polynomial time and, assuming truthful bids, achieves social welfare at least 50% of the maximum possible.[9]

You may have been lulled into complacency, thinking that every reasonable allocation rule is monotone. The only non-monotone rule that we've seen is the "second-highest bidder wins" rule for single-item auctions (Section 3.3), which we don't care about anyways. Consider yourself warned.

Warning

Natural allocation rules are not always monotone.

For example, for every $\epsilon > 0$, there is a $(1-\epsilon)$-approximation algorithm for the knapsack problem that runs in time polynomial in the input and $\frac{1}{\epsilon}$—a "fully polynomial-time approximation scheme (FPTAS)" (see Problem 4.2). The rule induced by the standard implementation of this algorithm is *not* monotone, although it can be tweaked to restore monotonicity without degrading the approximation guarantee (again, see Problem 4.2). This is characteristic of work in algorithmic mechanism design: for an $\mathcal{NP}$-hard optimization problem of interest, check if the state-of-the-art approximation

[9] The greedy allocation rule is even better under additional assumptions. For example, if $w_i \leq \alpha W$ for every bidder i, with $\alpha \in (0, \frac{1}{2}]$, then the approximation guarantee improves to $1-\alpha$, even if the third step of the algorithm is omitted.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

algorithm directly leads to a DSIC mechanism. If not, tweak it or design a new approximation algorithm that does, hopefully without degrading the approximation guarantee.

4.3 The Revelation Principle

4.3.1 DSIC Revisited

To this point, our mechanism design theory has studied only DSIC mechanisms. We reiterate that there are good reasons to strive for a DSIC guarantee. First, it is easy for a participant to figure out what to do in a DSIC mechanism: just play the obvious dominant strategy and truthfully reveal one's private information. Second, the designer can predict the mechanism's outcome assuming only that participants play their dominant strategies, a relatively weak behavioral assumption. Nevertheless, non-DSIC mechanisms like first-price auctions (Section 2.3) are also important in practice.

Can non-DSIC mechanisms accomplish things that DSIC mechanisms cannot? To answer this question, we need to tease apart two different conditions that are conflated in our DSIC definition (Definition 2.3).[10]

The DSIC Condition

(1) For every valuation profile, the mechanism has a *dominant-strategy equilibrium*—an outcome that results from every participant playing a dominant strategy.

(2) In this dominant-strategy equilibrium, every participant truthfully reports her private information to the mechanism.

There are mechanisms that satisfy (1) but not (2). To give a silly example, imagine a single-item auction in which the seller, given bids $\mathbf{b}$, runs a Vickrey auction on the bids $2\mathbf{b}$. Every bidder's dominant strategy is then to bid half her value.

[10] We'll ignore the "individual rationality" condition in Definition 2.3, which does not matter for the main points of this section.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

4.3.2 Justifying Direct Revelation

The revelation principle states that, given requirement (1) in Section 4.3.1, requirement (2) comes for free.

Theorem 4.3 (Revelation Principle for DSIC Mechanisms) *For every mechanism M in which every participant always has a dominant strategy, there is an equivalent direct-revelation DSIC mechanism M'.*

"Equivalence" in Theorem 4.3 means that, for every valuation profile, the outcome (e.g., winners of an auction and selling prices) of M' under direct revelation is identical to that of M when agents play their dominant strategies.

Proof: The proof uses a simulation argument; see Figure 4.2. By assumption, for every participant i and private information v_i that i might have, i has a dominant strategy $s_i(v_i)$ in the given mechanism M.

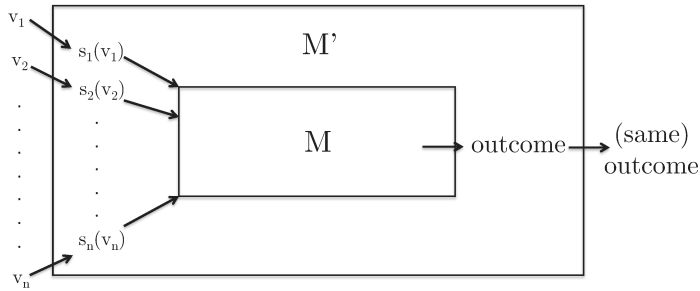


Figure 4.2: Proof of the revelation principle. Construction of the direct-revelation mechanism M', given a mechanism M with dominant strategies.

We next construct a mechanism M', to which participants delegate the responsibility of playing the appropriate dominant strategy. Precisely, the (direct-revelation) mechanism M' accepts bids $b_1, \ldots, b_n$ from the agents. It submits the bids $s_1(b_1), \ldots, s_n(b_n)$ to the mechanism M and chooses the same outcome that M does.

Mechanism M' is DSIC: If a participant i has private information v_i, then submitting a bid other than v_i can only result in M' playing a strategy other than $s_i(v_i)$ in M, which can only decrease i's utility. ■

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

The point of Theorem 4.3 is that, at least in principle, if you want to design a mechanism with dominant strategies, then you might as well design one in which direct revelation (in auctions, truthful bidding) is a dominant strategy. Thus *truthfulness per se is not important*; what makes DSIC mechanism design difficult is the requirement that a desired outcome is a dominant-strategy equilibrium.

4.3.3 Beyond Dominant-Strategy Equilibria

Can we obtain better mechanisms by relaxing condition (1) from Section 4.3.1? An immediate issue with this idea is that, when agents do not have dominant strategies, we require stronger behavioral assumptions to predict what participants will do and what the mechanism's outcome will be. For example, we can consider a Bayes-Nash equilibrium with respect to a common prior distribution over the private information of the participants (see Problem 5.3) or a Nash equilibrium in a full-information model (similar to Problem 3.1). If we're willing to make such assumptions, can we do better than with DSIC mechanisms?

The answer is "sometimes, yes." For this reason, and because non-DSIC mechanisms are common in practice, it is important to develop mechanism design theory beyond DSIC mechanisms. Remark 5.5 and Problem 5.3 offer a brief glimpse of this theory. A rough rule of thumb is that, for sufficiently simple problems like those we've studied up until now, DSIC mechanisms can do everything that non-DSIC mechanisms can. In more complex problems, however, weakening the DSIC constraint often allows the designer to achieve performance guarantees that are provably impossible for DSIC mechanisms. DSIC and non-DSIC mechanisms are incomparable in such settings—the former enjoy stronger incentive guarantees, the latter better performance guarantees. Which of these is more important depends on the details of the application.

The Upshot

☆ Knapsack auctions model the allocation of a shared resource with limited capacity. Bidders have private valuations and publicly known sizes. In a feasible outcome, the total size of

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

the winning bidders is at most the resource capacity.

☆ The problem of computing the outcome of a knapsack auction that maximizes social welfare is $\mathcal{NP}$-hard. Thus, if $\mathcal{P} \neq \mathcal{NP}$, there are no ideal knapsack auctions.

☆ The goal in algorithmic mechanism design is to relax the second requirement of ideal auctions (welfare maximization) as little as possible, subject to the first (DSIC) and third (polynomial-time) requirements. In the best-case scenario, there is a polynomial-time DSIC mechanism with an approximate welfare guarantee matching that of state-of-the-art polynomial-time approximation algorithms.

☆ State-of-the-art approximation algorithms for the welfare maximization problem may or may not induce monotone allocation rules.

☆ The revelation principle states that, for every mechanism with a dominant-strategy equilibrium, there is an equivalent mechanism in which direct revelation is a dominant-strategy equilibrium.

☆ In many complex mechanism design problems, non-DSIC mechanisms can achieve performance guarantees that are provably impossible for DSIC mechanisms.

Notes

The origins of algorithmic mechanism design are in Nisan and Ronen (2001) and Lehmann et al. (2002); see Nisan (2015) for a recent survey. Single-parameter environments are studied by Archer and Tardos (2001). Knapsack auctions are introduced in

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

Mu'Alem and Nisan (2008). The first formulation of the revelation principle appears in Gibbard (1973). Garey and Johnson (1979) give a good exposition of $\mathcal{NP}$-completeness and how to interpret it.

Problem 4.1 is related to Chekuri and Gamzu (2009). The classical FPTAS for the knapsack problem (Problem 4.2) is due to Ibarra and Kim (1975); see the books by Vazirani (2001) and Williamson and Shmoys (2010) for detailed coverage of this and dozens of other state-of-the-art polynomial-time approximation algorithms. The rest of Problem 4.2 is from Briest et al. (2005). Problem 4.3 is from Lehmann et al. (2002).

Exercises

Exercise 4.1 Consider an arbitrary single-parameter environment, with feasible set X. Prove that the welfare-maximizing allocation rule

$$\mathbf{x}(\mathbf{b}) = \text{argmax}_{(x_1,\ldots,x_n)\in X} \sum_{i=1}^{n} b_i x_i \tag{4.2}$$

is monotone in the sense of Definition 3.6.

[Assume that ties are broken in a deterministic and consistent way, such as lexicographically.]

Exercise 4.2 Continuing the previous exercise, restrict now to feasible sets X that contain only 0-1 vectors—that is, each bidder either wins or loses. We can identify each feasible outcome with a "feasible set" of bidders (the winners). Assume that for every bidder i, there is an outcome in which i does not win. Myerson's payment formula (3.5) dictates that a winning bidder pays her "critical bid"—the infimum of the bids at which she would continue to win.

Prove that, when S^* is the set of winning bidders under the allocation rule (4.2) and $i \in S^*$, i's critical bid equals the difference between (1) the maximum social welfare of a feasible set that excludes i; and (2) the social welfare $\sum_{j\in S^*\setminus\{i\}} v_j$ of the bidders other than i in the chosen outcome S^*.

[In this sense, each winning bidder pays her "externality"—the welfare loss she imposes on others.]

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

Exercise 4.3 Continuing the previous exercise, consider a 0-1 single-parameter environment. Suppose you are given a subroutine that, given bids $\mathbf{b}$, computes the outcome of the welfare-maximizing allocation rule (4.2).

(a) Explain how to implement a welfare-maximizing DSIC mechanism by invoking this subroutine $n + 1$ times, where n is the number of participants.

(b) Conclude that mechanisms that are ideal in the sense of Theorem 2.4 exist for precisely the families of single-parameter environments in which the welfare-maximization problem (given $\mathbf{b}$ as input, compute (4.2)) can be solved in polynomial time.

Exercise 4.4 Prove that the greedy algorithm in the proof of Theorem 4.2 always computes an optimal fractional knapsack solution.

Exercise 4.5 Prove that the three-step greedy knapsack auction allocation rule in Section 4.2.2 is monotone. Does it remain monotone with the two optimizations discussed in the footnotes?

Exercise 4.6 Consider a variant of a knapsack auction in which we have two knapsacks, with known capacities W_1 and W_2. Feasible sets of this single-parameter environment now correspond to subsets S of bidders that can be partitioned into sets S_1 and S_2 satisfying $\sum_{i \in S_j} w_i \leq W_j$ for $j = 1, 2$.

Consider the allocation rule that first uses the single-knapsack greedy allocation rule of Section 4.2.2 to pack the first knapsack, and then uses it again on the remaining bidders to pack the second knapsack. Does this algorithm define a monotone allocation rule? Give either a proof of this fact or an explicit counterexample.

Exercise 4.7 *(H)* The revelation principle (Theorem 4.3) states that (direct-revelation) DSIC mechanisms can simulate all other mechanisms with dominant-strategy equilibria. Critique the revelation principle from a practical perspective. Name a specific situation where you might prefer a non-direct-revelation mechanism with a dominant-strategy equilibrium to the corresponding DSIC mechanism, and explain your reasoning.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

Problems

Problem 4.1 Consider a variant of a knapsack auction in which both the valuation v_i and the size w_i of each bidder i are private. A mechanism now receives both bids $\mathbf{b}$ and reported sizes $\mathbf{a}$ from the bidders. An allocation rule $\mathbf{x}(\mathbf{b}, \mathbf{a})$ now specifies the amount of capacity allocated to each bidder, as a function of the bids and reported sizes. Feasibility dictates that $\sum_{i=1}^{n} x_i(\mathbf{b}, \mathbf{a}) \leq W$ for every $\mathbf{b}$ and $\mathbf{a}$, where W is the total capacity of the shared resource. We define the utility of a bidder i as $v_i - p_i(\mathbf{b}, \mathbf{a})$ if she gets her required capacity (i.e., $x_i(\mathbf{b}, \mathbf{a}) \geq w_i$) and as $-p_i(\mathbf{b}, \mathbf{a})$ otherwise. This is not a single-parameter environment.

Consider the following mechanism. Given bids $\mathbf{b}$ and reported sizes $\mathbf{a}$, the mechanism runs the greedy knapsack auction of Section 4.2.2, taking the reported sizes $\mathbf{a}$ at face value, to obtain a subset of winning bidders and prices $\mathbf{p}$. The mechanism concludes by awarding each winning bidder capacity equal to her reported size a_i, at a price of p_i; losing bidders receive and pay nothing. Is this mechanism DSIC? Prove it or give an explicit counterexample.

Problem 4.2 Section 4.2.2 gives an allocation rule for knapsack auctions that is monotone, guarantees at least 50% of the maximum social welfare, and runs in polynomial time. Can we do better?

We first describe a classical fully polynomial-time approximation scheme (FPTAS) for the knapsack problem. The input to the problem is item values $v_1, \ldots, v_n$, item sizes $w_1, \ldots, w_n$, and a knapsack capacity W. For a user-supplied parameter $\epsilon > 0$, we consider the following algorithm $\mathcal{A}_\epsilon$; m is a parameter that will be chosen shortly.

- Round each v_i up to the nearest multiple of m, call it v_i'.
- Divide the v_i''s through by m to obtain integers $\tilde{v}_1, \ldots, \tilde{v}_n$.
- For item values $\tilde{v}_1, \ldots, \tilde{v}_n$, compute the optimal solution using a pseudopolynomial-time algorithm.

  [You can assume that there exists such an algorithm with running time polynomial in n and $\max_{i=1}^{n} \tilde{v}_i$.]

(a) Prove that if we run algorithm $\mathcal{A}_\epsilon$ with the parameter m set to $\epsilon(\max_{i=1}^{n} v_i)/n$, then the running time of the algorithm is polynomial in n and $\frac{1}{\epsilon}$ (independent of the v_i's).

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

(b) *(H)* Prove that if we run algorithm $\mathcal{A}_\epsilon$ with the parameter m set to $\epsilon(\max_{i=1}^{n} v_i)/n$, then the algorithm outputs a solution with total value at least $1-\epsilon$ times the maximum possible.

(c) Prove that if we run algorithm $\mathcal{A}_\epsilon$ with the parameter m set to a fixed constant, independent of the v_i's, then the algorithm yields a monotone allocation rule.

(d) Prove that if we run algorithm $\mathcal{A}_\epsilon$ with the parameter m set as in (a) and (b), then the algorithm need *not* yield a monotone allocation rule.

(e) *(H)* Give a DSIC mechanism for knapsack auctions that, for a user-specified parameter ϵ and assuming truthful bids, outputs an outcome with social welfare at least $1-\epsilon$ times the maximum possible, in time polynomial in n and $\frac{1}{\epsilon}$.

Problem 4.3 Consider a set M of distinct items. There are n bidders, and each bidder i has a publicly known subset $T_i \subseteq M$ of items that it wants, and a private valuation v_i for getting them. If bidder i is awarded a set S_i of items at a total price of p, then her utility is $v_i x_i - p$, where x_i is 1 if $S_i \supseteq T_i$ and 0 otherwise. This is a single-parameter environment. Since each item can only be awarded to one bidder, a subset W of bidders can all receive their desired subsets simultaneously if and only if if $T_i \cap T_j = \emptyset$ for each distinct $i, j \in W$.

(a) *(H)* Prove that the problem of computing a welfare-maximizing feasible outcome, given the v_i's and T_i's as input, is $\mathcal{NP}$-hard.

(b) Here is a greedy algorithm for the social welfare maximization problem, given bids $\mathbf{b}$ from the bidders.

```
initialize W = ∅ and X = M
sort and re-index the bidders so that
b_1 ≥ b_2 ≥ ··· ≥ b_n
for i = 1, 2, 3, ..., n do
    if T_i ⊆ X then
        remove T_i from X and add i to W
return winning bidders W
```

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

Does this algorithm define a monotone allocation rule? Prove it or give an explicit counterexample.

(c) *(H)* Prove that if all bidders report truthfully and have sets T_i of cardinality at most d, then the outcome of the allocation rule in (b) has social welfare at least $\frac{1}{d}$ times that of the maximum possible.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.005

Lecture 5

Revenue-Maximizing Auctions

Lectures 2–4 focused on the design of mechanisms that maximize, exactly or approximately, the social welfare of the outcome. Revenue is generated in such mechanisms only as a side effect, a necessary evil to incentivize agents to report truthfully their private information. This lecture studies mechanisms that are designed to raise as much revenue as possible, and characterizes the expected revenue-maximizing mechanism with respect to a prior distribution over agents' valuations.

Section 5.1 explains why reasoning about revenue maximization is harder than welfare maximization, and introduces Bayesian environments. Section 5.2 is the heart of this lecture, and it characterizes expected revenue-maximizing mechanisms as "virtual welfare maximizers." Section 5.3 describes how this theory was used to boost sponsored search revenue at Yahoo. Section 5.4 proves a technical lemma needed for the characterization in Section 5.2.

5.1 The Challenge of Revenue Maximization

5.1.1 Spoiled by Social Welfare Maximization

There are several reasons to begin the study of mechanism design with the objective of maximizing social welfare. The first reason is that this objective is relevant to many real-world scenarios. For instance, in government auctions (e.g., to sell wireless spectrum; see Lecture 8), the primary objective is welfare maximization. Revenue is also a consideration in such auctions, but it is usually not the first-order objective. Also, in competitive markets, a rule of thumb is that a seller should focus on welfare maximization, since otherwise a competitor will (thereby stealing their customers).

The second reason to start with social welfare maximization is pedagogical: social welfare is special. In every single-parameter envi-

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

ronment, there is a DSIC mechanism that, for every profile of private valuations, assuming truthful bids, computes the welfare-maximizing outcome (cf., Exercise 4.1).[1] Such a mechanism optimizes the social welfare as effectively as if all of the private information was known in advance—the DSIC constraint is satisfied for free. This amazingly strong performance guarantee, called an "ex post" guarantee, cannot generally be achieved for other objective functions.

5.1.2 One Bidder and One Item

The following trivial example is illuminating. Suppose there is one item and only one bidder, with a private valuation v. With only one bidder, the space of direct-revelation DSIC auctions is small: they are precisely the *posted prices*, meaning take-it-or-leave-it offers.[2] With a posted price of $r \geq 0$, the auction's revenue is either r (if $v \geq r$) or 0 (if $v < r$).

Maximizing social welfare in this setting is trivial: just set $r = 0$, so that the auction always awards the item to the bidder for free. This optimal posted price is *independent of* v.

Suppose we wanted to maximize revenue. How should we set r? If we telepathically knew v, we would set $r = v$. But with v private to the bidder, what should we do? It is not clear how to reason about this question.

The fundamental issue is that the revenue-maximizing auction varies with the private valuations. With a single item and bidder, a posted price of 20 will do very well on inputs where v is 20 or a little larger, and terribly when v is less than 20 (while smaller posted prices will do better). Welfare maximization, for which there is an input-independent optimal DSIC mechanism, is special indeed.

5.1.3 Bayesian Analysis

To compare the revenue of two different auctions, we require a model to compare trade-offs across different inputs. The classical approach is to use *average-case* or *Bayesian* analysis. We consider a model comprising the following ingredients:

[1] This holds even more generally; see Lecture 7.

[2] These are the deterministic DSIC auctions. An auction can also randomize over posted prices, but the point of this example remains the same.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

- A single-parameter environment (Section 3.1). We assume that there is a constant M such that $x_i \leq M$ for every i and feasible solution $(x_1, \ldots, x_n) \in X$.

- Independent distributions $F_1, \ldots, F_n$ with positive and continuous density functions $f_1, \ldots, f_n$. We assume that the private valuation v_i of participant i is drawn from the distribution F_i.[3] We also assume that the support of every distribution F_i belongs to $[0, v_{\max}]$ for some $v_{\max} < \infty$.[4]

A key assumption is that the mechanism designer knows the distributions $F_1, \ldots, F_n$.[5] The realizations $v_1, \ldots, v_n$ of agents' valuations are private, as usual. Since we focus on DSIC auctions, where agents have dominant strategies, the agents do not need to know the distributions $F_1, \ldots, F_n$.[6]

In a Bayesian environment, it is clear how to define the "optimal" mechanism—it is the one that, among all DSIC mechanisms, has the highest expected revenue (assuming truthful bids). The expectation is with respect to the given distribution $F_1 \times F_2 \times \cdots \times F_n$ over valuation profiles.

5.1.4 One Bidder and One Item, Revisited

With our Bayesian model, single-bidder single-item auctions are easy to reason about. The expected revenue of a posted price r is simply

$$\underbrace{r}_{\text{revenue of a sale}} \cdot \underbrace{(1 - F(r))}_{\text{probability of a sale}}.$$

Given a distribution F, it is usually a simple matter to solve for the best posted price r. An optimal posted price is called a *monopoly*

[3] The distribution function $F_i(z)$ denotes the probability that a random variable with distribution F_i has value at most z.

[4] The results of this lecture hold more generally if every distribution F_i has finite expectation.

[5] In practice, these distributions are typically derived from data, such as bids in past auctions.

[6] In mechanisms without dominant strategies, such as first-price single-item auctions, the standard approach is to consider "Bayes-Nash equilibria"; see Problem 5.3 for details. Bayes-Nash equilibrium analysis assumes a "common prior," meaning that all of the agents know the distributions $F_1, \ldots, F_n$.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

price of the distribution F. Since DSIC mechanisms are posted prices (and distributions thereof), posting a monopoly price is a revenue-maximizing auction. For instance, if F is the uniform distribution on $[0,1]$, so that $F(x) = x$ on $[0,1]$, then the monopoly price is $\frac{1}{2}$, achieving an expected revenue of $\frac{1}{4}$.

5.1.5 Multiple Bidders

The plot thickens already with two bidders, where the space of DSIC auctions is more complicated than the space of posted prices. For example, consider a single-item auction with two bidders with valuations drawn independently from the uniform distribution on $[0,1]$. We could of course run a second-price auction (Section 2.4); its expected revenue is the expected value of the smaller bid, which is $\frac{1}{3}$ (Exercise 5.1(a)).

We can also supplement a second-price auction with a *reserve price*, analogous to the opening bid in an eBay auction. In a second-price auction with reserve r, the allocation rule awards the item to the highest bidder, unless all bids are less than r, in which case no one gets the item. The corresponding payment rule charges the winner (if any) the second-highest bid or r, whichever is larger. From a revenue standpoint, adding a reserve price r is both good and bad: you lose revenue when all bids are less than r, but when exactly one bid is above r the reserve price boosts the revenue. With two bidders with valuations drawn independently from the uniform distribution on $[0,1]$, adding a reserve price of $\frac{1}{2}$ raises the expected revenue of a second-price auction from $\frac{1}{3}$ to $\frac{5}{12}$ (Exercise 5.1(b)). Can we do better? Either by using a different reserve price, or with a entirely different auction format?

5.2 Characterization of Optimal DSIC Mechanisms

The primary goal of this lecture is to give an explicit description of an optimal (i.e., expected revenue-maximizing) DSIC mechanism for every single-parameter environment and distributions $F_1, \ldots, F_n$.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

5.2.1 Preliminaries

We can simplify the problem by applying the revelation principle from last lecture (Theorem 4.3). Since every DSIC mechanism is equivalent to—and hence has the same expected revenue as—a direct-revelation DSIC mechanism $(\mathbf{x}, \mathbf{p})$, we can restrict our attention to such mechanisms. We correspondingly assume truthful bids (i.e., $\mathbf{b} = \mathbf{v}$) for the rest of the lecture.

The expected revenue of a DSIC mechanism $(\mathbf{x}, \mathbf{p})$ is, by definition,

$$\mathbf{E}_{\mathbf{v} \sim \mathbf{F}} \left[\sum_{i=1}^{n} p_i(\mathbf{v}) \right], \tag{5.1}$$

where the expectation is with respect to the distribution $\mathbf{F} = F_1 \times \cdots \times F_n$ over agents' valuations. It is not clear how to directly maximize the expression (5.1) over the space of DSIC mechanisms. We next work toward a *second* formula for the expected revenue of a mechanism. This alternative formula only references the allocation rule of a mechanism, and not its payment rule, and for this reason is far easier to maximize.

5.2.2 Virtual Valuations

The second formula for expected revenue uses the important concept of virtual valuations. For an agent i with valuation distribution F_i and valuation v_i, her *virtual valuation* is defined as

$$\varphi_i(v_i) = v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}. \tag{5.2}$$

The virtual valuation of an agent depends on her own valuation and distribution, and not on those of the other agents. For example, if F_i is the uniform distribution on $[0, 1]$, with $F_i(z) = z$ for $z \in [0, 1]$, then $f_i(z) = 1$, and $\varphi_i(z) = z - \frac{1-z}{1} = 2z - 1$ on $[0, 1]$. A virtual valuation is always at most the corresponding valuation, and it can be negative. See Exercise 5.2 for more examples.

Virtual valuations play a central role in the design of expected revenue-maximizing auctions. But what do they mean? One way to

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

interpret the formula

$$\varphi_i(v_i) = \underbrace{v_i}_{\text{what you'd like to charge}} - \underbrace{\frac{1 - F_i(v_i)}{f_i(v_i)}}_{\text{information rent earned by agent}}$$

is to think of v_i as the maximum revenue obtainable from agent i, and the second term as the inevitable revenue loss caused by not knowing v_i in advance, known as the *information rent*. A second interpretation of $\varphi_i(v_i)$ is as the slope of a "revenue curve" at v_i, where the revenue curve plots the expected revenue obtained from an agent with valuation drawn from F_i as a function of the probability of a sale. Problem 5.1 elaborates on this interpretation.

5.2.3 Expected Revenue Equals Expected Virtual Welfare

The following lemma is the workhorse of our characterization of optimal auctions. We give the proof, which is really just some calculus, in Section 5.4.

Lemma 5.1 *For every single-parameter environment with valuation distributions $F_1, \ldots, F_n$, every DSIC mechanism $(\mathbf{x}, \mathbf{p})$, every agent i, and every value $\mathbf{v}_{-i}$ of the valuations of the other agents,*

$$\mathbf{E}_{v_i \sim F_i}[p_i(\mathbf{v})] = \mathbf{E}_{v_i \sim F_i}[\varphi_i(v_i) \cdot x_i(\mathbf{v})]\,. \tag{5.3}$$

That is, the expected payment of an agent equals the expected virtual value earned by the agent. This identity holds only in expectation over v_i, and not pointwise.[7]

Taking Lemma 5.1 as given, we have the following important result.

Theorem 5.2 (Exp. Revenue Equals Exp. Virtual Welfare) *For every single-parameter environment with valuation distributions $F_1, \ldots, F_n$ and every DSIC mechanism $(\mathbf{x}, \mathbf{p})$,*

$$\underbrace{\mathbf{E}_{\mathbf{v} \sim \mathbf{F}}\left[\sum_{i=1}^{n} p_i(\mathbf{v})\right]}_{\text{expected revenue}} = \underbrace{\mathbf{E}_{\mathbf{v} \sim \mathbf{F}}\left[\sum_{i=1}^{n} \varphi_i(v_i) \cdot x_i(\mathbf{v})\right]}_{\text{expected virtual welfare}}. \tag{5.4}$$

[7] For example, virtual valuations can be negative while payments are always nonnegative.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

Proof: Taking the expectation, with respect to $\mathbf{v}_{-i} \sim \mathbf{F}_{-i}$, of both sides of (5.3) we obtain

$$\mathbf{E}_{\mathbf{v}\sim\mathbf{F}}[p_i(\mathbf{v})] = \mathbf{E}_{\mathbf{v}\sim\mathbf{F}}[\varphi_i(v_i) \cdot x_i(\mathbf{v})]\,.$$

Applying the linearity of expectation (twice) then gives

$$\begin{aligned}\mathbf{E}_{\mathbf{v}\sim\mathbf{F}}\left[\sum_{i=1}^{n} p_i(\mathbf{v})\right] &= \sum_{i=1}^{n} \mathbf{E}_{\mathbf{v}\sim\mathbf{F}}[p_i(\mathbf{v})] \\ &= \sum_{i=1}^{n} \mathbf{E}_{\mathbf{v}\sim\mathbf{F}}[\varphi_i(v_i) \cdot x_i(\mathbf{v})] \\ &= \mathbf{E}_{\mathbf{v}\sim\mathbf{F}}\left[\sum_{i=1}^{n} \varphi_i(v_i) \cdot x_i(\mathbf{v})\right],\end{aligned}$$

as desired. ■

The second term in (5.4) is our second formula for the expected revenue of a mechanism, and we should be pleased with its simplicity. If we replaced the $\varphi_i(v_i)$'s by v_i's, then we would be left with an old friend: the expected *welfare* of the mechanism. For this reason, we refer to $\sum_{i=1}^{n} \varphi_i(v_i) \cdot x_i(\mathbf{v})$ as the *virtual welfare* of a mechanism on the valuation profile $\mathbf{v}$. Theorem 5.2 implies that maximizing expected revenue over the space of DSIC mechanisms reduces to maximizing expected virtual welfare over the same space.

5.2.4 Maximizing Expected Virtual Welfare

It is shocking that a formula as simple as (5.4) holds. It says that even though we only care about payments, we can focus on an optimization problem that concerns only the mechanism's allocation rule. This second form is far more operational, and we proceed to determine the mechanisms that maximize it.

How should we choose the allocation rule $\mathbf{x}$ to maximize the expected virtual welfare

$$\mathbf{E}_{\mathbf{v}\sim\mathbf{F}}\left[\sum_{i=1}^{n} \varphi_i(v_i) x_i(\mathbf{v})\right]? \tag{5.5}$$

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

We have the freedom of choosing $\mathbf{x}(\mathbf{v})$ for each valuation profile $\mathbf{v}$, and have no control over the input distribution $\mathbf{F}$ or the virtual values $\varphi_i(v_i)$. Thus, the obvious approach is to *maximize pointwise*: separately for each $\mathbf{v}$, we choose $\mathbf{x}(\mathbf{v})$ to maximize the virtual welfare $\sum_{i=1}^{n} \varphi_i(v_i)x_i(\mathbf{v})$ obtained on the input $\mathbf{v}$, subject to feasibility of the allocation. We call this the *virtual welfare-maximizing allocation rule*. This is the same as the welfare-maximizing allocation rule of (4.1) and (4.2), except with agents' valuations replaced by their virtual valuations (5.2).

For example, in a single-item auction, where the feasibility constraint is $\sum_{i=1}^{n} x_i(\mathbf{v}) \leq 1$ for every $\mathbf{v}$, the virtual welfare-maximizing rule just awards the item to the bidder with the highest virtual valuation. Well, not quite: recall that virtual valuations can be negative—for instance, $\varphi_i(v_i) = 2v_i - 1$ when v_i is uniformly distributed between 0 and 1—and if every bidder has a negative virtual valuation, then the virtual welfare is maximized by not awarding the item to anyone.[8]

The virtual welfare-maximizing allocation rule maximizes the expected virtual welfare (5.5) over *all* allocation rules, monotone or not. The key question is: *Is the virtual welfare-maximizing rule monotone?* If so, then by Myerson's lemma (Theorem 3.7) it can be extended to a DSIC mechanism, and by Theorem 5.2 this mechanism has the maximum possible expected revenue.

5.2.5 Regular Distributions

Monotonicity of the virtual welfare-maximizing allocation rule depends on the valuation distributions. The next definition identifies a sufficient condition for monotonicity.

Definition 5.3 (Regular Distribution) A distribution F is *regular* if the corresponding virtual valuation function $v - \frac{1-F(v)}{f(v)}$ is nondecreasing.

If every agent's valuation is drawn from a regular distribution, then with consistent tie-breaking, the virtual welfare-maximizing allocation rule is monotone (Exercise 5.5).

[8] Recall from the single-bidder example in Section 5.1 that maximizing expected revenue entails not always selling the item.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

For example, the uniform distribution on $[0, 1]$ is regular because the corresponding virtual valuation function is $2v - 1$. Many other common distributions are also regular (Exercise 5.3). Irregular distributions include many multi-modal distributions and distributions with extremely heavy tails.

With regular valuation distributions, we can extend the (monotone) virtual welfare-maximizing allocation rule to a DSIC mechanism using Myerson's lemma (Theorem 3.7). This is an expected revenue-maximizing DSIC mechanism.[9]

Virtual Welfare Maximizer

Assumption: the valuation distribution F_i of every agent is regular (Definition 5.3).

1. Transform the (truthfully reported) valuation v_i of agent i into the corresponding virtual valuation $\varphi_i(v_i)$ according to (5.2).

2. Choose the feasible allocation $(x_1, \ldots, x_n)$ that maximizes the virtual welfare $\sum_{i=1}^{n} \varphi_i(v_i)x_i$.[10]

3. Charge payments according to Myerson's payment formula (see (3.5) and (3.6)).[11]

We call this mechanism the *virtual welfare maximizer* for the given single-parameter environment and valuation distributions.

Theorem 5.4 (Virtual Welfare Maximizers Are Optimal) *For every single-parameter environment and regular distributions $F_1, \ldots, F_n$, the corresponding virtual welfare maximizer is a DSIC mechanism with the maximum-possible expected revenue.*

A stunning implication of Theorem 5.4 is that revenue-maximizing mechanisms are almost the same as welfare-maximizing

[9] With additional work, the results of this lecture can be extended to irregular valuation distributions. See the Notes for details.

[10] The simplest way to break ties is lexicographically with respect to some fixed total ordering over the feasible outcomes.

[11] If every x_i can only be 0 or 1, then these payments are particularly simple: every winner pays the infimum of the bids at which she would continue to win, holding others' bids fixed.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

mechanisms, and differ only in using virtual valuations in place of valuations. In this sense, revenue maximization reduces to welfare maximization.

Remark 5.5 (Bayesian Incentive Compatible Mechanisms) Generalizing the derivations in Section 3.4 and this section yields a substantially stronger version of Theorem 5.4: the mechanism identified in the theorem maximizes expected revenue not only over all DSIC mechanisms but more generally over all "Bayesian incentive compatible (BIC)" mechanisms. A BIC mechanism for valuation distributions $F_1, \ldots, F_n$ is one in which truthful revelation forms a Bayes-Nash equilibrium (see Problem 5.3 for a definition). Every DSIC mechanism is BIC with respect to every choice of $F_1, \ldots, F_n$. Since optimizing expected revenue over all BIC mechanisms yields a DSIC mechanism, the DSIC property comes for free. The revelation principle (Theorem 4.3) can be adapted to BIC mechanisms (Problem 5.4), implying that, under the assumptions of Theorem 5.4, no Bayes-Nash equilibrium of any mechanism (e.g., first-price auctions) results in expected revenue larger than that earned by the optimal DSIC mechanism.

5.2.6 Optimal Single-Item Auctions

Theorem 5.4 gives a satisfying solution to the problem of expected revenue-maximizing mechanism design, in the form of a relatively explicit and easy-to-implement optimal mechanism. However, these mechanisms are not easy to interpret. Do they ever simplify to familiar mechanisms?

Let's return to single-item auctions. Assume that bidders are *i.i.d.*, meaning that they have a common valuation distribution F and hence a common virtual valuation function φ. Assume also that F is *strictly regular*, meaning that φ is a strictly increasing function. The virtual-welfare-maximizing mechanism awards the item to the bidder with the highest nonnegative virtual valuation, if any. Since all bidders share the same increasing virtual valuation function, the bidder with the highest virtual valuation is also the bidder with the highest valuation. This allocation rule is the same as that of a second-price auction with a reserve price of $\varphi^{-1}(0)$. By Theorem 3.7(b), the payment rules also coincide. Thus, for any number of i.i.d. bidders and a strictly regular valuation distribution, eBay (with a suitable

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

opening bid) is the optimal auction format! Returning to the setting described at the end of Section 5.1, with all valuations distributed uniformly on $[0, 1]$, the second-price auction with reserve $\frac{1}{2} = \varphi^{-1}(0)$ is optimal. Given the richness of the DSIC auction design space, it is astonishing that such a simple and practical auction pops out as the theoretically optimal one.

5.3 Case Study: Reserve Prices in Sponsored Search

So how does all this optimal mechanism design theory get used, anyway? This section discusses a 2008 field experiment that explored whether or not the lessons of optimal auction theory could be used to increase sponsored search revenue at Yahoo.

Recall from Section 2.6 our model of sponsored search auctions. Which such auction maximizes the expected revenue, at least in theory? If we assume that bidders' valuations-per-click are drawn i.i.d. from a regular distribution F with virtual valuation function φ, then the optimal auction considers only bidders who bid at least the reserve price $\varphi^{-1}(0)$, and ranks these bidders by bid (from the best slot to the worst). See Exercise 5.8.

What had Yahoo been doing, up to 2008? First, they were using relatively low reserve prices—initially \$.01, later \$.05, and then \$.10. Perhaps more naively, they were using the same reserve price of \$.10 across all keywords, even though some keywords surely warranted higher reserve prices than did others (e.g., "divorce lawyer" versus "pizza"). How would Yahoo's revenue change if reserve prices were updated, independently for each keyword, to the theoretically optimal ones?

In the first step of the field experiment, a lognormal valuation distribution was fitted to past bidding data for approximately 500,000 different keywords.[12] The qualitative conclusions of the experiment appear to be independent of the details of this step, such as the particular family of valuation distributions chosen.

[12] Since Yahoo, like other search engines, uses a non-DSIC auction based on the GSP auction (Problem 3.1), one cannot expect the bids to be truthful. In this field experiment, valuations were reversed engineered from bids under the assumption that bidders are playing the equilibrium that is outcome-equivalent to the dominant-strategy outcome of the revenue-maximizing DSIC sponsored search auction (Exercise 5.8).

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

In the second step, theoretically optimal reserve prices were computed for each keyword, assuming that valuations were distributed according to the fitted distributions. As expected, the optimal reserve price varied significantly across keywords. There were plenty of keywords with a theoretically optimal reserve price of \$.30 or \$.40. Yahoo's uniform reserve price was much too low, relative to the advice provided by optimal auction theory, on these keywords.

The obvious experiment is to try out the theoretically optimal (and generally higher) reserve prices to see how they do. Yahoo's top brass wanted to be a little more conservative, though, and set the new reserve prices to be the average of the old ones (\$.10) and the theoretically optimal ones.[13] And the change worked: auction revenues went up several percent (of a very large number). The new reserve prices were especially effective in markets that are valuable but "thin," meaning not very competitive (less than six bidders). Better reserve prices were credited by Yahoo's president as the biggest reason for higher search revenue in Yahoo's third-quarter report in 2008.

*5.4 Proof of Lemma 5.1

This section sketches a proof of Lemma 5.1, that the expected (over $v_i \sim F_i$) revenue obtained from an agent i equals the expected virtual value that she earns. As a starting point, recall Myerson's payment formula (3.6)

$$p_i(v_i, \mathbf{v}_{-i}) = \int_0^{v_i} z \cdot x_i'(z, \mathbf{v}_{-i}) dz$$

for the payment made by agent i in a DSIC mechanism with allocation rule $\mathbf{x}$ on the valuation profile $\mathbf{v}$. We derived this equation assuming that the allocation function $x_i(z, \mathbf{v}_{-i})$ is differentiable. By standard advanced calculus, the same formula holds more generally for an arbitrary monotone function $x_i(z, \mathbf{v}_{-i})$, including piecewise constant functions, for a suitable interpretation of the derivative $x_i'(z, \mathbf{v}_{-i})$ and the corresponding integral. Similarly, all of the following proof

[13] Both in theory and empirically, this more conservative change accounts for most of the revenue increase. There are usually diminishing returns to revenue as the reserve price approaches the theoretical optimum, providing flexibility near the optimal price. The intuition for this principle is that the derivative of the expected revenue with respect to the reserve price is 0 at the optimal point.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

steps, which make use of calculus maneuvers like integration by parts, can be made fully rigorous for arbitrary bounded monotone functions without significant difficulty. We leave the details to the interested reader.[14]

Equation (3.6) states that payments are fully dictated by the allocation rule. Thus, at least in principle, we can express the expected revenue of an auction purely in terms of its allocation rule, with no explicit reference to its payment rule. Will the resulting revenue formula be easier to maximize than the original one? It's hard to know without actually doing it, so let's do it.

Step 1: Fix an agent i. By Myerson's payment formula, we can write the expected (over $v_i \sim F_i$) payment by i for a given value of $\mathbf{v}_{-i}$ as

$$\begin{aligned}\mathbf{E}_{v_i \sim F_i}[p_i(\mathbf{v})] &= \int_0^{v_{\max}} p_i(\mathbf{v}) f_i(v_i) dv_i \\ &= \int_0^{v_{\max}} \left[\int_0^{v_i} z \cdot x_i'(z, \mathbf{v}_{-i}) dz \right] f_i(v_i) dv_i.\end{aligned}$$

The first equality exploits the independence of agents' valuations—the fixed value of $\mathbf{v}_{-i}$ has no bearing on the distribution F_i from which v_i is drawn.

This step is exactly what we knew was possible in principle—rewriting the expected payment in terms of the allocation rule. For this to be useful, we need some simplifications.

Step 2: Whenever you have a double integral (or double sum) that you don't know how to interpret, it's worth reversing the integration order. Reversing the order of integration in

$$\int_0^{v_{\max}} \left[\int_0^{v_i} z \cdot x_i'(z, \mathbf{v}_{-i}) dz \right] f_i(v_i) dv_i$$

yields

$$\int_0^{v_{\max}} \left[\int_z^{v_{\max}} f_i(v_i) dv_i \right] z \cdot x_i'(z, \mathbf{v}_{-i}) dz,$$

which simplifies to

$$\int_0^{v_{\max}} (1 - F_i(z)) \cdot z \cdot x_i'(z, \mathbf{v}_{-i}) dz,$$

[14] For example, every bounded monotone function is integrable, and is differentiable except at a countable set of points.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

suggesting that we're on the right track.

Step 3: Integration by parts is also worth trying when massaging an integral into a more interpretable form, especially if there's an obvious derivative hiding in the integrand. We again get some encouraging simplifications:

$$\begin{aligned}
&\int_0^{v_{\max}} \underbrace{(1-F_i(z))\cdot z}_{g(z)} \cdot \underbrace{x_i'(z,\mathbf{v}_{-i})}_{h'(z)}\, dz \\
&= \underbrace{(1-F_i(z))\cdot z\cdot x_i(z,\mathbf{v}_{-i})\big|_0^{v_{\max}}}_{=0-0} \\
&\qquad - \int_0^{v_{\max}} x_i(z,\mathbf{v}_{-i})\cdot(1-F_i(z)-zf_i(z))dz \\
&= \int_0^{v_{\max}} \underbrace{\left(z-\frac{1-F_i(z)}{f_i(z)}\right)}_{=\varphi_i(z)} x_i(z,\mathbf{v}_{-i})f_i(z)dz. \qquad (5.6)
\end{aligned}$$

Step 4: We can interpret (5.6) as an expected value, with z drawn from the distribution F_i. Recalling the definition (5.2) of virtual valuations, this expectation is $\mathbf{E}_{v_i\sim F_i}[\varphi_i(v_i)\cdot x_i(\mathbf{v})]$. Summarizing, we have

$$\mathbf{E}_{v_i\sim F_i}[p_i(\mathbf{v})] = \mathbf{E}_{v_i\sim F_i}[\varphi_i(v_i)\cdot x_i(\mathbf{v})]\,,$$

as desired.

The Upshot

- ☆ Unlike welfare-maximizing mechanisms, the revenue-maximizing mechanism for an environment varies with the (private) valuations.
- ☆ In the average-case or Bayesian approach to comparing different mechanisms, each agent's valuation is drawn independently from a distribution known to the mechanism designer. The optimal mechanism is the one with the highest expected revenue with respect to these distributions.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

- ☆ The expected revenue of a DSIC mechanism can be expressed purely in terms of its allocation rule, using the important concept of virtual valuations (5.2).
- ☆ A distribution is regular if the corresponding virtual valuation function is nondecreasing. Many common distributions are regular.
- ☆ With regular valuation distributions, the optimal mechanism is a virtual welfare maximizer, which for each valuation profile chooses an outcome with maximum virtual welfare.
- ☆ In a single-item auction with bidders' valuations drawn i.i.d. from a regular distribution, the optimal auction is a second-price auction with a reserve price.
- ☆ The lessons learned from the theory of optimal mechanism design were used in 2008 to increase Yahoo's sponsored search revenue by several percent.

Notes

The model and main results of this lecture are due to Myerson (1981), as are the mentioned extensions to irregular valuation distributions and to Bayesian incentive compatible mechanisms (Remark 5.5). Myerson (1981) also notes the crucial importance of the independence assumption on agents' valuations, an observation that is developed further by Crémer and McLean (1985). With irregular distributions, the virtual welfare-maximizing allocation rule is not monotone, and it is necessary to solve for the monotone allocation rule with the maximum expected virtual welfare. This can be done by "ironing" virtual valuation functions to make them monotone, while at the same time preserving the virtual welfare of the mechanisms that matter. See Hartline (2016) for a textbook treatment of these extensions.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

The field experiment with reserve prices in Yahoo sponsored search auctions (Section 5.3) is reported by Ostrovsky and Schwarz (2009). The revenue curve interpretation of virtual valuations in Problem 5.1 is due to Bulow and Roberts (1989). Problem 5.2 is from Azar et al. (2013). Problem 5.3 is closely related to the property of "revenue equivalence," identified already in Vickrey (1961); see Krishna (2010) for an excellent exposition.

Exercises

Exercise 5.1 Consider a single-item auction with two bidders with valuations drawn independently from the uniform distribution on $[0, 1]$.

(a) Prove that the expected revenue obtained by a second-price auction (with no reserve) is $\frac{1}{3}$.

(b) Prove that the expected revenue obtained by a second-price auction with reserve $\frac{1}{2}$ is $\frac{5}{12}$.

Exercise 5.2 Compute the virtual valuation function of the following distributions.

(a) The uniform distribution on $[0, a]$ with $a > 0$.

(b) The exponential distribution with rate $\lambda > 0$ (on $[0, \infty)$).

(c) The distribution given by $F(v) = 1 - \frac{1}{(v+1)^c}$ on $[0, \infty)$, where $c > 0$ is some constant.

Exercise 5.3 Which of the distributions in Exercise 5.2 are regular (Definition 5.3)?

Exercise 5.4 A valuation distribution meets the *monotone hazard rate (MHR)* condition if its *hazard rate* $\frac{f_i(v_i)}{1-F_i(v_i)}$ is nondecreasing in v_i.[15]

[15] For intuition behind the MHR condition, consider waiting for a light bulb to burn out. Given that the bulb hasn't burned out yet, the probability that it burns out right now is increasing in the amount of time that has elapsed.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

(a) Prove that every distribution meeting the MHR condition is regular.

(b) Which of the distributions in Exercise 5.2 satisfy the MHR condition?

Exercise 5.5 Prove that for every single-parameter environment and regular valuation distributions $F_1, \ldots, F_n$, the virtual-welfare-maximizing allocation rule is monotone in the sense of Definition 3.6. Assume that ties are broken lexicographically with respect to some fixed total ordering over the feasible outcomes.

Exercise 5.6 *(H)* For the valuation distribution in Exercise 5.2(c), with $c = 1$, argue that the expected revenue of an auction does *not* necessarily equal its expected virtual welfare. How do you reconcile this observation with Theorem 5.2?

Exercise 5.7 Consider a k-unit auction (Example 3.2) in which bidders' valuations are drawn i.i.d. from a regular distribution F. Describe an optimal auction. Which of the following does the reserve price depend on: k, n, or F?

Exercise 5.8 Repeat the previous exercise for sponsored search auctions (Example 3.3).

Exercise 5.9 Consider a single-parameter environment and regular valuation distributions $F_1, \ldots, F_n$. For $\alpha \in [0, 1]$, call a DSIC mechanism an *α-approximate virtual welfare maximizer* if it always selects a feasible allocation with virtual welfare at least α times the maximum possible. Prove that the expected revenue of an α-approximate virtual welfare maximizer is at least α times that of an optimal mechanism.

Exercise 5.10 In the sponsored search auction case study in Section 5.3, raising reserve prices was particularly effective for valuable keywords (typical valuations-per-click well above the old reserve price of \$.10) that had few bidders (6 or less). Give at least two examples of keywords that you think might have these properties, and explain your reasoning.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

Problems

Problem 5.1 This problem derives an interesting interpretation of a virtual valuation $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ and the regularity condition. Consider a strictly increasing distribution function F with a strictly positive density function f on the interval $[0, v_{\max}]$, with $v_{\max} < +\infty$.

For a single bidder with valuation drawn from F, for $q \in [0,1]$, define $V(q) = F^{-1}(1-q)$ as the (unique) posted price that yields a probability q of a sale. Define $R(q) = q \cdot V(q)$ as the expected revenue obtained from a single bidder when the probability of a sale is q. The function $R(q)$, for $q \in [0,1]$, is the *revenue curve* of F. Note that $R(0) = R(1) = 0$.

(a) What is the revenue curve for the uniform distribution on $[0,1]$?

(b) Prove that the slope of the revenue curve at q (i.e., $R'(q)$) is precisely $\varphi(V(q))$, where φ is the virtual valuation function for F.

(c) Prove that a distribution is regular if and only if its revenue curve is concave.

Problem 5.2 *(H)* Consider a single bidder with valuation drawn from a regular distribution F that satisfies the assumptions of Problem 5.1. Let p be the *median* of F, meaning the value for which $F(p) = \frac{1}{2}$. Prove that the price p earns at least 50% of the expected revenue of the optimal posted price for F.

Problem 5.3 This problem introduces the Bayes-Nash equilibrium concept and compares the expected revenue earned by first- and second-price single-item auctions.

First-price auctions have no dominant strategies, and we require a new concept to reason about them. For this problem, assume that bidders' valuations are drawn i.i.d. from a commonly known distribution F. A *strategy* of a bidder i in a first-price auction is a predetermined plan for bidding—a function $b_i(\cdot)$ that maps her valuation v_i to a bid $b_i(v_i)$. The semantics are: "when my valuation is v_i, I will bid $b_i(v_i)$." We assume that bidders' strategies are common knowledge, with bidders' valuations (and hence induced bids) private as usual.

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006

A strategy profile $b_1(\cdot), \cdots, b_n(\cdot)$ is a *Bayes-Nash equilibrium* if every bidder always bids optimally given her information—if for every bidder i and every valuation v_i, the bid $b_i(v_i)$ maximizes i's expected utility, where the expectation is with respect to the distribution over others bids induced by F and $\mathbf{b}_{-i}$.

(a) Suppose F is the uniform distribution on $[0, 1]$. Verify that setting $b_i(v_i) = v_i(n-1)/n$ for every i and v_i yields a Bayes-Nash equilibrium.

(b) Prove that the expected revenue of the seller (over $\mathbf{v}$) at this equilibrium of the first-price auction is exactly the expected revenue of the seller in the truthful outcome of a second-price auction.

(c) *(H)* Extend the conclusion in (b) to every continuous and strictly increasing distribution function F on $[0, 1]$.

Problem 5.4 This problem uses first-price auctions to illustrate the extension of the revelation principle (Theorem 4.3) to Bayesian incentive compatible mechanisms (Remark 5.5).

(a) Let $F_1, \ldots, F_n$ be valuation distributions and $b_1, \ldots, b_n$ a Bayes-Nash equilibrium of a first-price auction, as defined in Problem 5.3. Prove that there exists a single-item auction M' such that truthful bidding is a Bayes-Nash equilibrium and, for every valuation profile $\mathbf{v}$, the truthful outcome of M' is identical to the equilibrium outcome of the first-price auction.

(b) A first-price auction is "prior-independent" in that its description makes no reference to bidders' valuation distributions. (See also Section 6.4.) Is the auction M' in part (a) prior-independent in this sense?

Downloaded from https://www.cambridge.org/core. AMS/Kaist Digital Sci Lib, on 16 Jul 2018 at 12:17:19, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781316779309.006